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PI Name:	Gupta, Ashwani D.Sc.		
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PI Email:	akgupta@eng.umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-5276
Organization Name:	University of Maryland		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:			
PI Web Page:			
City:	College Park	**State:**	MD
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Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
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The characteristics of gaseous diffusion flames have been obtained using high temperature combustion air under microgravity and normal gravity conditions. The time resolved flame images under free fall microgravity conditions were obtained from the video images obtained. The tests results reported here were conducted using propane as the fuel and about 1000oC combustion air. The burner included a central fuel jet injected into the surrounding high temperature combustion air. The fuel jet exit Reynolds number was 63 with a nozzle diameter of 0.686 mm. Several measurements were taken at different air preheats and fuel jet exit Reynolds number. The resulting hybrid color flame was found to be blue at the base of the flame followed by a yellow color flame. The length and width of flame during the entire free fall conditions has been examined. Also the relative flame length and width for blue and yellow portion of the flame has been examined under microgravity conditions. The results show that the flame length decreases and width increases with high air preheats in microgravity condition. In microgravity conditions the flame length is larger with normal temperature

Task Description:

combustion air than high temperature air. Correlations used to predict the shapes of round non-buoyant laminar flames have been used to examine the shapes of high temperature air flames under microgravity conditions. The results show favorable predictions for the flame width during the microgravity conditions.

Introduction

Study of laminar diffusion flames has important role to understand and utilize their thermal and chemical behavior in practical combustion systems. Modeling of laminar diffusion flames has somewhat less complication and provides a convenient comparison with the experimental data. The flame length and width is an important property of laminar flames. Many investigators have examined the flame length and shapes. Past measurements of laminar jet diffusion flame shapes under microgravity conditions have been carried out using either drop towers to provide microgravity condition or aircraft facilities to provide long duration low gravity condition. But all of these experiments have been carried out using normal temperature combustion air. Cochran and co-worker [1-3] studied the hydrocarbon-fueled laminar jet diffusion flames using a 2.2 sec free fall drop tower studies. This was then followed by several studies by Cochran and co-workers and Bahadori and co-workers [4-9] using both 2.2 sec and 5.2 sec. drop towers. They tried to minimize the problem of transient behavior of flame during the free fall.

Sunderland et al. [10] measured the shape of laminar jet diffusion flames using the 2.2 sec drop tower facility. In contrast some theoretical studies have also been carried out to develop the empirical models and to evaluate the theoretical prediction on the shapes of laminar jet diffusion flames. Several laminar diffusion flame models, of varying complexity, have also been proposed [11-17] that successfully predict the linear correlation between stoichiometric flame length and the fuel flow rate. Among these, the analysis of Spalding [13] offers a potentially simple and robust method for estimating the shapes of steady nonbuoyant round laminar jet diffusion flames. Lin et al [18], based on the Spalding [13] simplified theoretical analysis, proposed a correlation for nonbuoyant laminar jet diffusion flames. Based on their correlation for luminous flame boundaries of nonbuoyant laminar jet diffusion flames, luminous flame lengths were found to increase linearly with the fuel flow rate but was relatively independent of the jet exit diameter and pressure. Maximum luminous flame diameters were found to increase linearly with jet exit diameter but were relatively independent of the fuel flow rate and pressure.

No previous data are available on the effect of combustion air preheat in microgravity conditions.

Combustion with high temperature air is a relatively new concept and has many practical benefits [19]. Some of the benefits of using high temperature air for combustion that have been demonstrated include (i) energy savings of about 30%, (ii) pollution reduction, including NOx and CO2, by about 30%, (iii) 25% reduction in the size of the equipment, and (iv) far uniform thermal field in the combustion zone. Limited data are available under normal gravity conditions [19-21]. High temperature air combustion is a relatively new concept for better combustion and heat transfer technology, and its applications are enormous including different kinds of furnaces for various industrial use and combustion engines [19]. Combustion with highly preheated combustion air has very complex kinetics and heat transfer characteristics. Jagnow [22] used direct flame photographic technique and other diagnostic techniques to provide qualitative and quantitative information on the characteristics of diffusion flames under normal gravity (1g) conditions using combustion air preheated to very high temperatures. No data has been reported in the open literature on the characteristics of diffusion flames formed with high temperature combustion air in microgravity conditions. In this study we provide information on the behavior of flames formed with high temperature combustion air in microgravity conditions. The results on flame length and width are complemented with simple available correlations to describe the general flame behavior.

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Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

High temperature air combustion technology has been accepted and now utilized by many countries around the world for significant energy savings, pollution reduction, and better quality of product produced from the process since the temperature in the entire combustion zone is very uniform (better than 40degrees C). Ultra low NOx, and carbon monoxide are produced with negligible unburnt hydrocarbons. Such achievements are new and provides a basis for many new applications.

Task Progress:

Experimental
The experiments were conducted using a 1 sec drop tower facility. The experimental burner consisted of a 0.686 mm diameter fuel jet that was centrally located in a 12.7 mm diameter air jet. The combustion air in the outer jet could be maintained at any temperature, up to about 1100oC using electrical heating. The experiment test rig consisted of burner, voltage regulator, electrical air preheater, combustion chamber, color CCD camera and digital camcorder, and gas cylinders for fuel, combustion air, and burner cooling. The flame images were captured using a CCD camera in the emission mode at a framing rate of 30 frames per second and then recorded onto a digital camcorder. Several experiments were conducted under microgravity conditions at normal pressure and normal oxygen concentration air using propane as the fuel at a flow rate 0.02 SCFH. The flame photographs, recorded with the color CCD camera and camcorder, were analyzed to determine the flame shape (length and width) and color.

Results and Discussion

The results are presented at two air preheat temperatures of 900oC and 1000oC. The experiments were conducted with combustion air preheated to temperatures in the range of 900oC (for experiment #3) to 1000oC (experiment #4) at a propane gas flow rate of 0.02 SCFH. Preparation for the experiments involved turning on the air preheater to get the desired air preheats temperature. In this paper results from two specific experiments (experiments #3 & 4) are reported to provide the effect of air preheat temperature on flame size, shape and color (hybrid) under microgravity and normal gravity conditions. Before the drop the video camera was turned on to record the flame features at 1-g. The time sequence photos of flame during drop were separated from the recorded images onto the video camera at a speed of 30 frames/s. The results show that in both cases the flames are of hybrid color consisting of initial blue color flame followed by a yellow color flame downstream of the burner. The length of the yellow color flame is higher at higher air preheat temperature (compare flame photos from experiments #4 with #3). The yellow portion of the flame with experiments # 4 (having higher air preheat temperature) is higher. A close examination of the entire flame length (blue and yellow portion) shows that the length of flame in experiment #4 is shorter as compared to flames produced in experiment #3. The flames photographs were further analyzed with respect to the total length and width as well as the length and width of the blue and yellow portion of the flames during the entire time duration in the microgravity environment. The results show shorter flame at higher air preheats temperatures. Immediately after the onset of drop (very low microgravity conditions) the flame length for experiment #4 (with high air preheats) is smaller. This suggests that higher temperature air reduces the flame length in microgravity.

Our results obtained at 1-g conditions also showed that the flame length decreases with increase in air preheats [20, 21] using normal air. In contrast the flame width increases with increase in air preheats under microgravity conditions (compare flame width from t =0 to about t = 0.1 for the two cases). At the onset of microgravity the flame with high air preheats temperature experiences a rapid increase in flame width (experiment #4) as compared to low temperature air case (experiment #3). The rate of increase of flame width is higher with higher air preheats at t =0.1 to about t = 0.2. After t = 0.2 sec both flames reach the maximum width due to lower gravity field. The change in gravity with time during the drop was calculated by balancing the force on the entire drop package (i.e., drag and gravity = mass x acceleration).

At higher residence times (t) under microgravity conditions the flame length increases because of the drag and hence a higher gravity values as compared to that obtained at the onset of drop. This suggests that under microgravity conditions the flames are smaller and wider with increase in air preheats. At higher air preheats the yellow portion of the flame becomes higher and the blue portion of the flame becomes smaller. Hotter flames (with higher air preheats) affect the shape more (with regard to their length and width) in the presence of microgravity. These results are in agreement with the studies carried out at normal gravity conditions. The fuel flow rate and temperature of the air calculated results was taken as 0.02 ft3 /hr. (i.e., same flow rate as in the measurements). While the trends for the flame width are similar, the predicted length of the flame is higher. This shows that high temperature air has significant effect on reducing the flame length. This observation has also been made in 1-g experiments. The relative change in flame width and length during the microgravity test duration is consistent with the change in microgravity conditions during the drop, i.e., smaller length and larger width with increase in microgravity during the drop. Global features on the change in flame length and width as affected by gravity and air preheat temperature show that high air preheats further widens the flame width and reduces the flame length in microgravity conditions.

Conclusions

1. Flame length decreases and the width increases under microgravity conditions.

2. At higher air preheat temperatures the flame length decreases under microgravity conditions. Under microgravity conditions at high air preheat temperature, the flame width is even wider as compared to normal temperature air and normal gravity conditions.

3. The flame luminosity as measured by the yellow color of the flame increases with increase in air preheats temperature

	under microgravity conditions. This is consistent with the findings at the normal gravity conditions. 4. The correlations provide reasonably good agreement on the flame width at high air preheats with that observed under microgravity conditions. 5. The data obtained provides useful information for model validation and model development. Acknowledgments This research was supported by NASA Microgravity Combustion Research Program, Program Scientist Dr. Merrill K. King. This support is gratefully acknowledged. The support provided by Dr. Dennis Stocker is much appreciated. References 20. Gupta, A. K.: Flame Characteristics and Challenges with High Temperature Air Combustion, Invited Lecture, Proc. 2nd International Seminar on High Temperature Combustion in Industrial Furnaces, Stockholm, Sweden, January 17-18, 2000. 21. Gupta, A. K.: Thermal Characteristics of Gaseous Fuel Flames using High Temperature Air, ASME J. Engineering for Gas Turbine and Power, Vol. 126, No. 1, January/February 2004, p. 9-19.
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